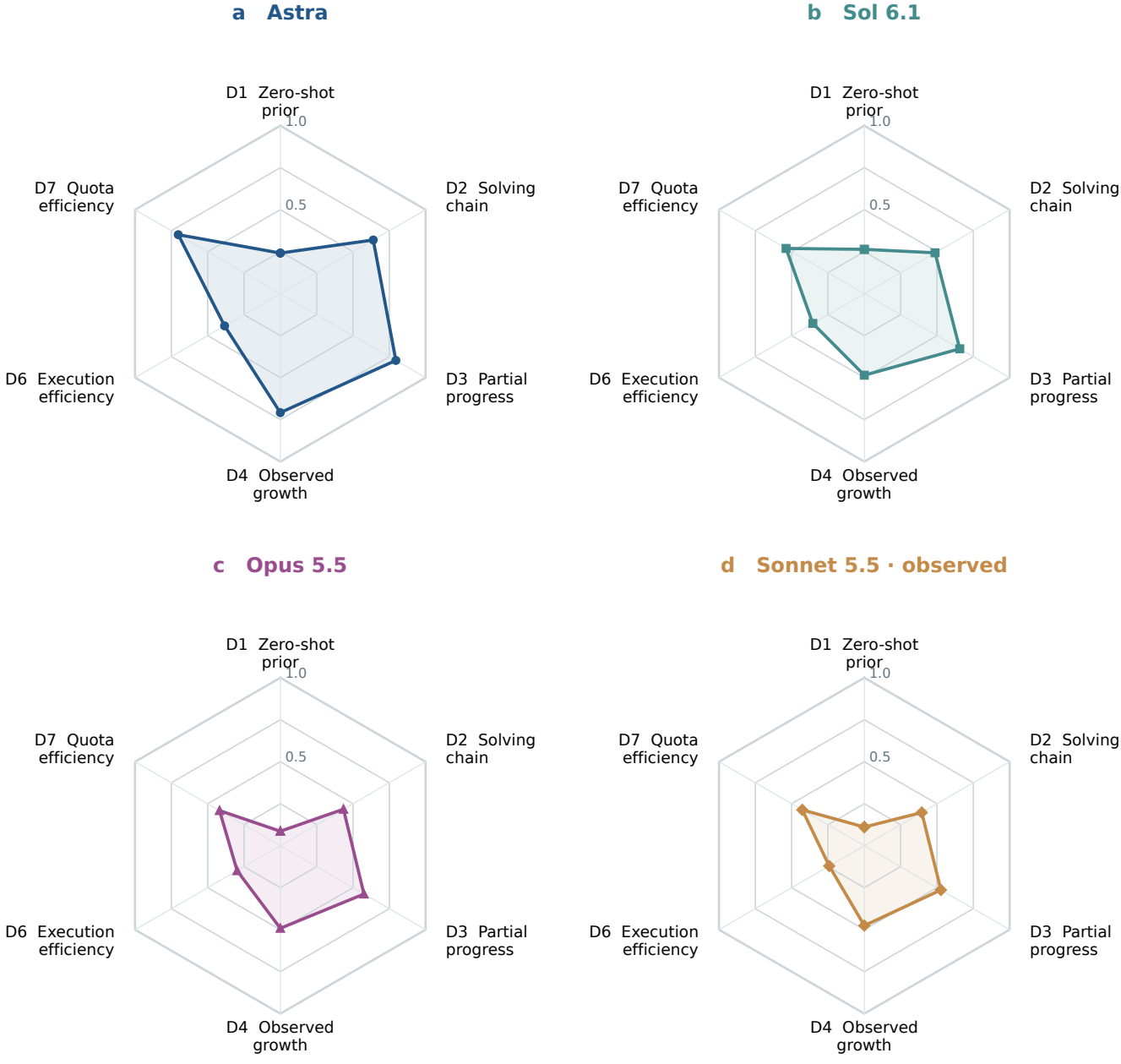
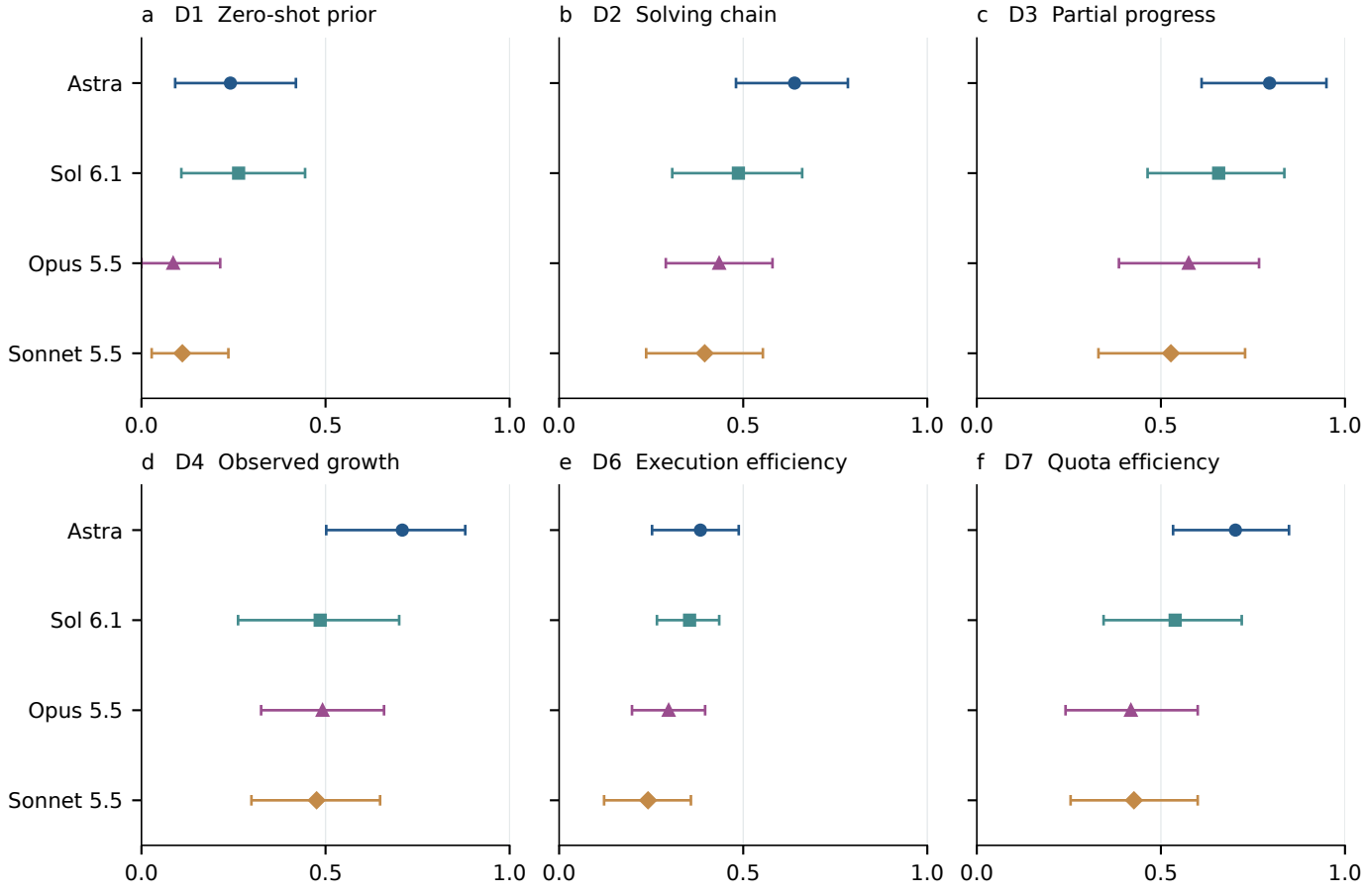
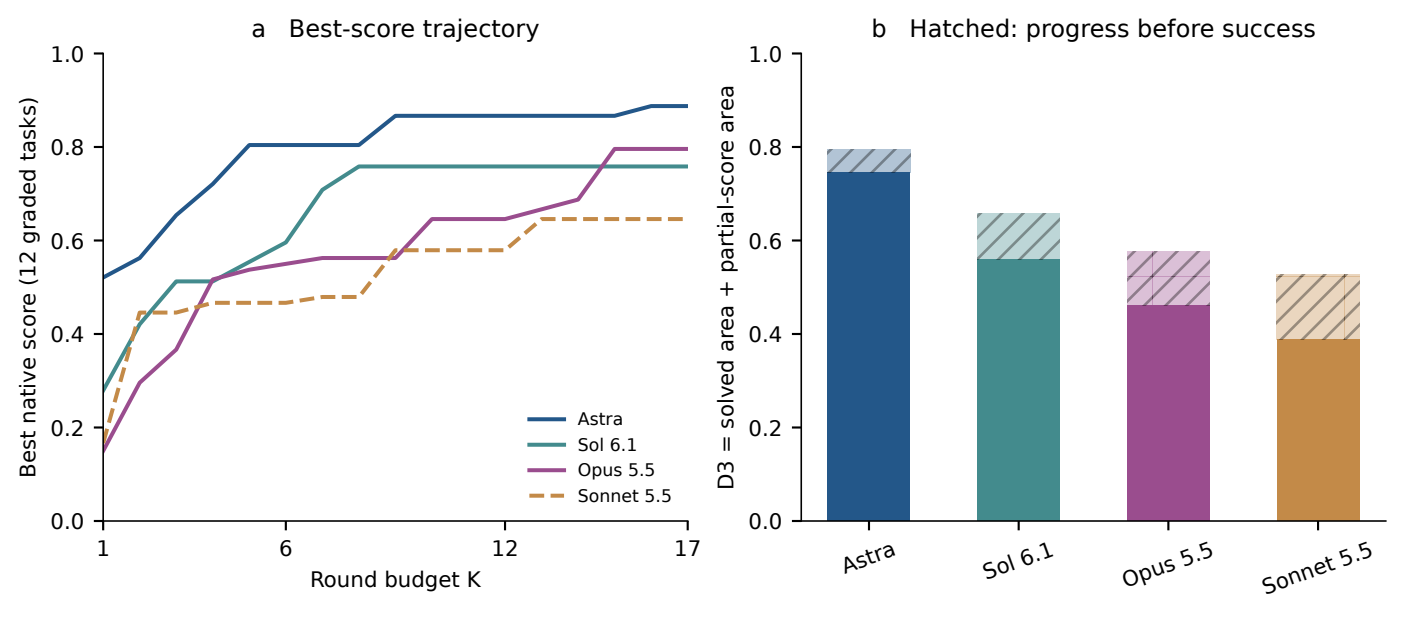
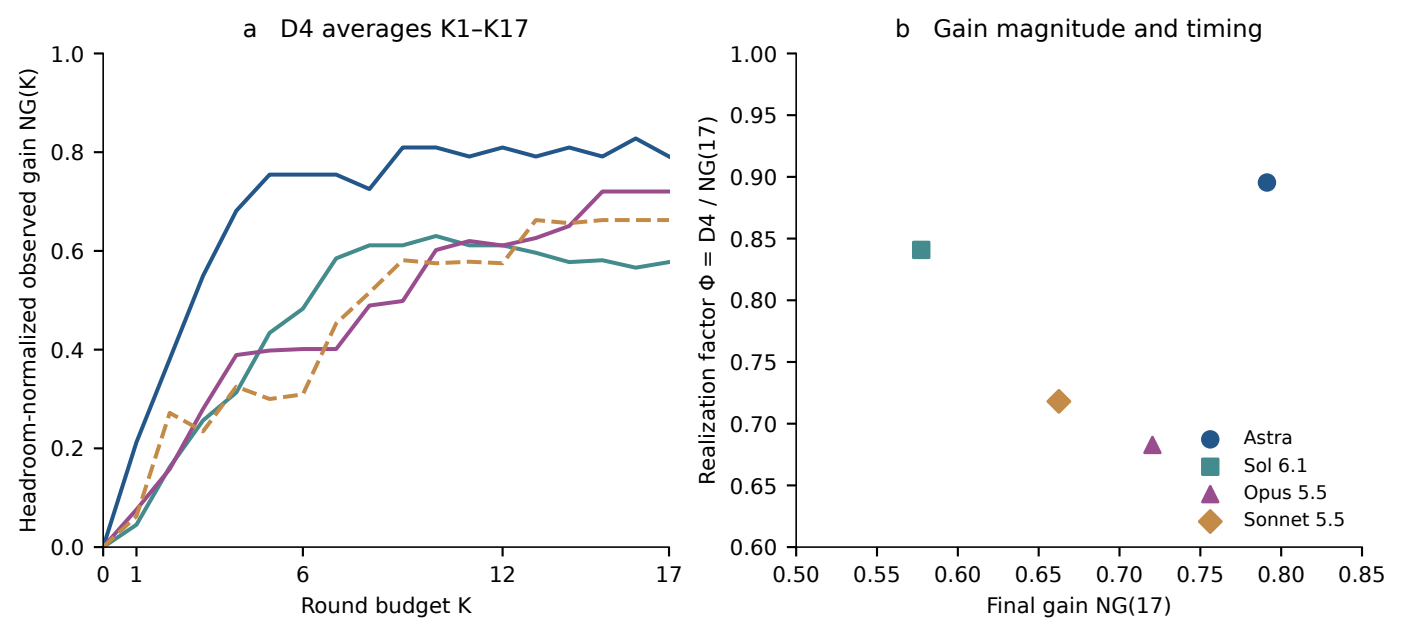


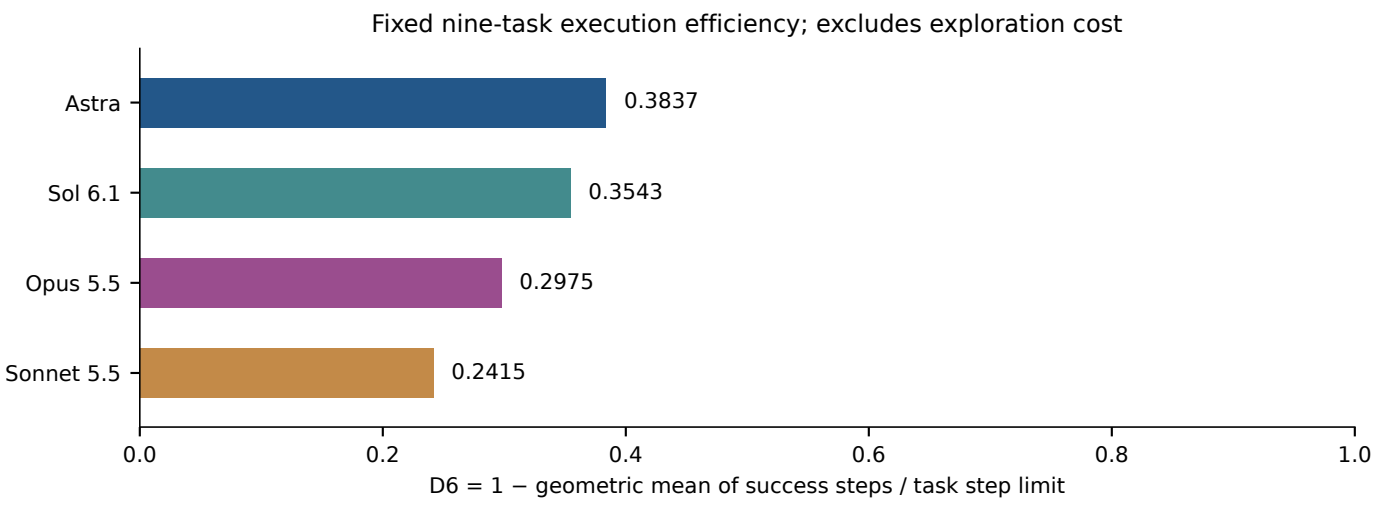
Fixed [0, 1] axes; higher = better by definition. D1/D2/D4/D7: 18 tasks; D3: 12 graded; D6: 9 fixed tasks.
Axes share evidence. Polygon size is not an independent capability measure. Sonnet remains incomplete.

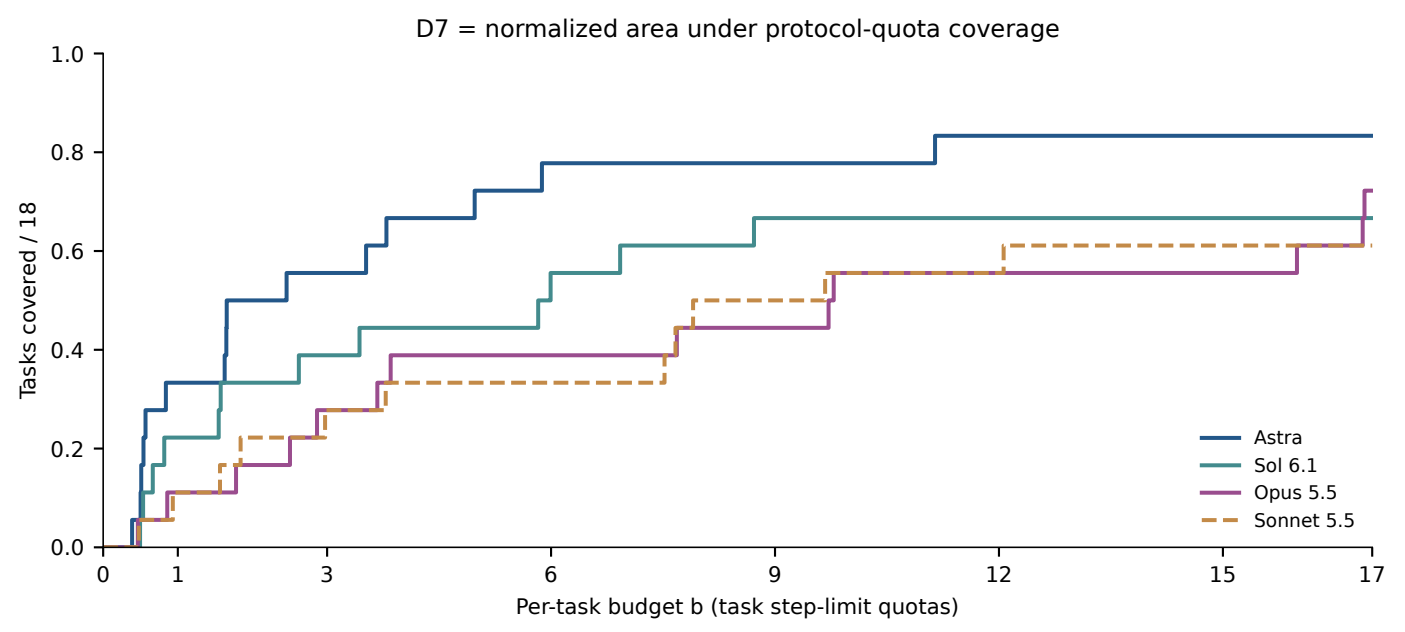


95% percentile intervals: joint paired task bootstrap, B = 5,000; seed = 20261012.
Conditional on observed runs; excludes layout, protocol and missing-completion uncertainty.


a Best-score trajectory
Best native score (12 graded tasks)
Round budget K
1.0
0.8
0.6
0.4
0.2
0.0
1
6
12
17
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
b Hatched: progress before success
D3 = solved area + partial-score area
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5


a D4 averages K1–K17
Headroom-normalized observed gain NG(K)
1.0
0.8
0.6
0.4
0.2
0.0
0
1
6
12
17
Round budget K
b Gain magnitude and timing
Realization factor Φ = D4 / NG(17)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
Final gain NG(17)
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5


Fixed nine-task execution efficiency; excludes exploration cost
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
0.3837
0.3543
0.2975
0.2415
0.0
0.2
0.4
0.6
0.8
1.0
D6 = 1 − geometric mean of success steps / task step limit



Formal steps + reasoning-timeout top-ups + one quota per recorded extra attempt.
This is protocol accounting, not measured physical consumption; unknown attempts remain unknown.